

A counterexample on the equality in the higher order Suita's conjecture

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1 Introduction

Let Ω be a planar domain in $\mathbb{C}$ bounded by n analytic Jordan curves: $\Gamma_1, \dots, \Gamma_n$, where we set Γ_n to be the boundary of the unbounded component of $\mathbb{C} \setminus \Omega$. According to the solvability of the Dirichlet problem on planar domains, for each $j \in \{1, \dots, n-1\}$, there exists a harmonic function u_j on $\overline{\Omega}$ such that

$$u_j|_{\Gamma_j} \equiv 1, \text{ and } u_j|_{\Gamma_k} \equiv 0, \quad \forall k \in \{1, \dots, n\} \setminus \{j\}.$$

In the present paper, we prove

THEOREM 1.1. *There exists a planar domain Ω in $\mathbb{C}$ bounded by 4 analytic Jordan curves, such that for the corresponding harmonic functions u_j , $j = 1, 2, 3$, there does not exist $z \in \Omega$ such that*

$$u_j(z) \in \mathbb{Q}, \quad j = 1, 2, 3.$$

The relevance to the higher order Suita conjecture is the following known characterization of Guan–Sun–Yuan [4]: for a planar domain bounded by n analytic Jordan curves, the equality in the m -th order Suita conjecture holds at z if and only if

$$(m+1)u_j(z) \in \mathbb{Z}, \quad j = 1, \dots, n-1.$$

Thus, if all $u_j(z)$ are rational, then with $m+1$ a common denominator of the $u_j(z)$ the equality holds at z ; conversely any such equality forces $u_j(z) \in \mathbb{Q}$ for all j . Consequently Theorem 1.1 provides a four-connected domain at no point of which the equality in the higher order Suita conjecture holds.

2 The counterexample and the proof

We prove Theorem 1.1 by a Baire-category (generic parameter) construction. No numerical computation is used; the only transcendental ingredient is the choice of a parameter outside a countable union of null sets.

Fix three pairwise distinct points $c_1, c_2, c_3 \in \mathbb{D}$ and choose $\rho_0 > 0$ so small that the closed disks $\overline{\mathbb{D}(c_j, 2\rho_0)}$ are pairwise disjoint and contained in $\mathbb{D}$. Let

$$B := \left(\frac{1}{2}\rho_0, \rho_0\right) \times \left(\frac{1}{2}\rho_0, \rho_0\right) \times \left(\frac{1}{2}\rho_0, \rho_0\right) \subset \mathbb{R}^3,$$

where $(\frac{1}{2}\rho_0, \rho_0)$ denotes the open interval in $\mathbb{R}$. For $\mathbf{r} = (r_1, r_2, r_3) \in B$ put

$$D_j(\mathbf{r}) := \overline{\mathbb{D}(c_j, r_j)}, \quad \Gamma_j(\mathbf{r}) := \partial D_j(\mathbf{r}), \quad j = 1, 2, 3, \quad \Gamma_4 := \partial \mathbb{D},$$

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and

$$\Omega(\mathbf{r}) := \mathbb{D} \setminus \left(D_1(\mathbf{r}) \cup D_2(\mathbf{r}) \cup D_3(\mathbf{r}) \right).$$

Then $\Omega(\mathbf{r})$ is bounded by the four pairwise disjoint analytic Jordan curves $\Gamma_1(\mathbf{r}), \Gamma_2(\mathbf{r}), \Gamma_3(\mathbf{r}), \Gamma_4$ (see Figure 1).

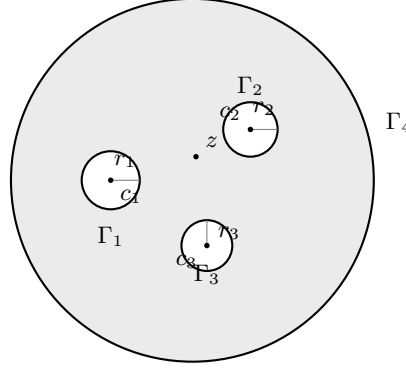


Figure 1: The four-connected domain $\Omega(\mathbf{r}) = \mathbb{D} \setminus (D_1(\mathbf{r}) \cup D_2(\mathbf{r}) \cup D_3(\mathbf{r}))$ with boundary components $\Gamma_1, \Gamma_2, \Gamma_3$ and Γ_4 .

For $j = 1, 2, 3$ let $u_j(\cdot; \mathbf{r})$ be the harmonic function on $\Omega(\mathbf{r})$ with boundary values 1 on $\Gamma_j(\mathbf{r})$ and 0 on the other three curves, and set $u_4 := 1 - u_1 - u_2 - u_3$. Basic facts on harmonic measure used below can be found in [2].

LEMMA 2.1. *For every $z \in \Omega(\mathbf{r})$ and $j, k \in \{1, 2, 3\}$,*

$$\frac{\partial u_j}{\partial r_j}(z; \mathbf{r}) > 0, \quad \frac{\partial u_j}{\partial r_k}(z; \mathbf{r}) < 0 \quad (j \neq k), \quad \frac{\partial u_4}{\partial r_k}(z; \mathbf{r}) < 0. \quad (2.1)$$

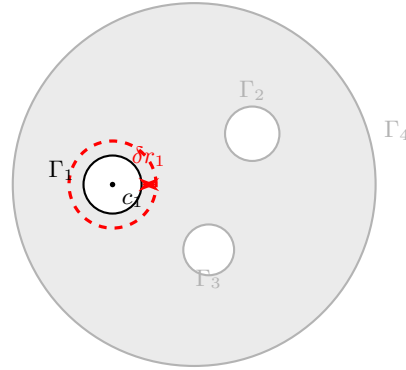


Figure 2: Enlargement of the boundary component $\Gamma_1(\mathbf{r})$ by $\delta r_1 > 0$: the dashed circle is the new boundary, and the red arrow indicates the outward normal displacement used in the variational formula (2.2).

Proof. The functions u_j are real-analytic in $(z, \mathbf{r})$; this follows from the standard analytic dependence of solutions of the Dirichlet problem on analytic boundary data [1]. The first inequality in (2.1) is the usual monotonicity of harmonic measure under enlargement of the corresponding boundary component (Figure 2); it follows from the maximum principle, since on the enlarged domain the old solution is < 1 on the new boundary and $= 0$ on all other boundary components.

For $k \neq j$ (and for u_4 with any k) we use Hadamard's variational formula. Let $G_{\mathbf{r}}(w, z)$ be the Green function of $\Omega(\mathbf{r})$ and let n denote the outward unit normal. Up to a fixed positive multiplicative constant,

$$\frac{\partial u_j}{\partial r_k}(z; \mathbf{r}) = \int_{\Gamma_k(\mathbf{r})} \frac{\partial u_j}{\partial n}(w) \frac{\partial G_{\mathbf{r}}}{\partial n}(w, z) ds(w). \quad (2.2)$$

For an inner boundary $\Gamma_k(\mathbf{r})$ one has $\partial G_{\mathbf{r}}/\partial n > 0$ on $\Gamma_k(\mathbf{r})$. By Hopf's lemma, $\partial u_j/\partial n$ on $\Gamma_k(\mathbf{r})$ is strictly positive if $j = k$ and strictly negative if $j \neq k$; similarly $\partial u_4/\partial n < 0$ on $\Gamma_k(\mathbf{r})$. Hence (2.1) follows from (2.2). $\blacksquare$

For $z \in \Omega(\mathbf{r})$ put

$$A(z; \mathbf{r}) := \left(\frac{\partial u_j}{\partial r_k}(z; \mathbf{r}) \right)_{1 \leq j, k \leq 3}.$$

LEMMA 2.2. *For every $\mathbf{r} \in B$ and every $z \in \Omega(\mathbf{r})$ the matrix $A(z; \mathbf{r})$ is invertible.*

Proof. By (2.1), $A_{jk} < 0$ for $j \neq k$, and

$$(A^T \mathbf{1})_j = \sum_{i=1}^3 \frac{\partial u_i}{\partial r_j} = \frac{\partial}{\partial r_j} (u_1 + u_2 + u_3) = -\frac{\partial u_4}{\partial r_j} > 0.$$

Thus A^T has positive diagonal, nonpositive off-diagonal entries and positive row sums. If $A^T \mathbf{x} = 0$ for some $\mathbf{x} \neq 0$, choose an index i at which $x_i = \max_j x_j > 0$; if all $x_j \leq 0$, choose instead an index i at which $x_i = \min_j x_j < 0$. In the first case, since $A_{ik}^T \leq 0$ for $k \neq i$,

$$0 = (A^T \mathbf{x})_i = A_{ii}^T x_i + \sum_{k \neq i} A_{ik}^T x_k \geq A_{ii}^T x_i + \sum_{k \neq i} A_{ik}^T x_i = (A^T \mathbf{1})_i x_i > 0,$$

a contradiction. The second case is analogous with all inequalities reversed. Hence A^T , and therefore A , is nonsingular. $\blacksquare$

Proof of Theorem 1.1. Let

$$X := \{(z, \mathbf{r}) \in \mathbb{C} \times B : z \in \Omega(\mathbf{r})\},$$

an open subset of $\mathbb{R}^2 \times \mathbb{R}^3 \cong \mathbb{R}^5$, and define

$$F : X \rightarrow \mathbb{R}^3, \quad F(z, \mathbf{r}) := (u_1(z; \mathbf{r}), u_2(z; \mathbf{r}), u_3(z; \mathbf{r})).$$

By Lemma 2.1, F is real-analytic. Its derivative contains the invertible 3×3 block $D_{\mathbf{r}} F = A(z; \mathbf{r})$; hence F is a submersion at every point of X .

For $\mathbf{q} = (q_1, q_2, q_3) \in \mathbb{Q}^3$ set $S_{\mathbf{q}} := F^{-1}(\mathbf{q})$. By the implicit function theorem, $S_{\mathbf{q}}$ is a real-analytic submanifold of X of dimension $\dim_{\mathbb{R}} X - 3 = 2$ (possibly empty). Let $\pi_{\mathbf{q}} : S_{\mathbf{q}} \rightarrow B$ be

the projection onto the parameter box. Each connected component of $S_{\mathbf{q}}$ is a two-dimensional C^1 manifold. Since $\dim_{\mathbb{R}} B = 3$, Sard's theorem shows that its image under $\pi_{\mathbf{q}}$ has empty interior (indeed Lebesgue measure zero). A second countable manifold has at most countably many connected components, so $\pi_{\mathbf{q}}(S_{\mathbf{q}})$ is a countable union of null sets, hence a null set. Consequently

$$E := \bigcup_{\mathbf{q} \in \mathbb{Q}^3} \pi_{\mathbf{q}}(S_{\mathbf{q}})$$

is a countable union of null sets, and $B \setminus E$ is nonempty (in fact of full measure).

Choose $\mathbf{r} \in B \setminus E$. We claim that $\Omega := \Omega(\mathbf{r})$ has the required property. Indeed, if for some $z \in \Omega$ all three numbers $u_j(z; \mathbf{r})$ were rational, then with $\mathbf{q} := (u_1(z; \mathbf{r}), u_2(z; \mathbf{r}), u_3(z; \mathbf{r})) \in \mathbb{Q}^3$ we would have $(z, \mathbf{r}) \in S_{\mathbf{q}}$ and hence $\mathbf{r} \in E$, a contradiction. This completes the proof. ■

REMARK 2.3. *The proof process of this counterexample comes from Codex combined with DeepSeek v4 flash.*

References

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